

SIMATS ENGINEERING ASSESSMENT TOOL

COURSE NAME: Theory of Computation

COURSE CODE: CSA13

COURSE OUTCOME COVERED

CO1: Analyze fundamental mathematical concepts of automata theory for formal languages and decision problems.(BL:3)

Assessment Tool Weightage: 40%

QUESTIONS: 5
Duration : 1 Hour

Total Marks:25

Analytical Problem

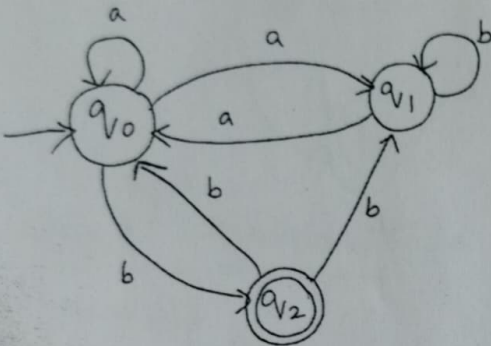
Code of Conduct:

Shauk Hafeeza Taramym (192372044)
(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

SIMATS ENGINEERING ASSESSMENT TOOL

1. Consider the NFA given below which accepts some language L over the alphabet $\Sigma = \{a, b\}$. Construct a DFA that represents the same language L

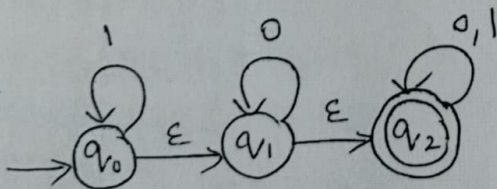


2. Let L be the language of words of length atleast 2 whose last two letters are different from each other. For example $abaab$ is in L but not bb , Construct an NFA for L , Then draw the equivalent DFA

3. Design a Nondeterministic Finite Automata (NFA) that takes a binary string as input and accepts the string only if the last two digits are the same (either 00 or 11). Write the formal definition of the NFA. Also show that the string **01011** is accepted by the NFA

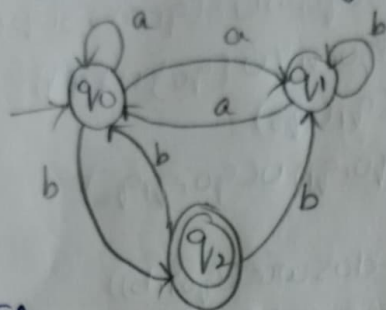
4. Consider the NFA given below which accepts the language representing the set of all strings having 001 as a substring over the alphabet $\Sigma = \{0, 1\}$. Construct a DFA that represents the same language.

5. A NFA with ϵ -transitions is given below. Find ϵ -closure for each state and Design an equivalent DFA which accepts the same language. Draw its transition table and mark the initial and final states



Consider the NFA given below which accepts same language L over the alphabet $\Sigma = \{a, b\}$. Construct a DFA that represents the same Language L ?

(A) Step 1:



NFA
Step 2: Transition table

State/ip	a	b
$\rightarrow q_0$	$\{q_0, q_1\}$	$\{q_2\}$
q_1	$\{q_0\}$	$\{q_1\}$
$* q_2$	$\{q_0\}$	$\{q_0, q_1\}$

Step 3:

Formula: $M = \{Q, \Sigma, \delta, q_0, F\}$

$$Q = \{q_0, q_1, q_2\}$$

$$\Sigma = \{a, b\}$$

$$S = \{q_0\}$$

$$F = \{q_2\}$$

$$\delta(q_0, a) = \{q_0, q_1\}$$

$$\delta(q_0, b) = \{q_2\}$$

$$\delta(q_1, a) = \{q_0\}$$

$$\delta(q_1, b) = \{q_1\}$$

$$\delta(q_2, a) = \emptyset$$

$$\delta(q_2, b) = \{q_0, q_1\}$$

Step 4:

$$E\text{-closure}(q_0) = \{q_0, q_1, q_2\}$$

$$E\text{-closure}(q_1) = \{q_0, q_1\}$$

$$E\text{-closure}(q_2) = \{q_0, q_1, q_2\}$$

$$\begin{aligned} \text{Steps: } S'(q_0, a) &= E\text{-closure}(\delta(E\text{-closure}(q_0, a), a)) \\ &= E\text{-closure}(\delta(q_0, q_1, q_2), a) \\ &= E\text{-closure}(\delta(q_0, a) \cup \delta(q_1, a) \cup \delta(q_2, a)) \\ &= E\text{-closure}(\{q_0, q_1\} \cup \{q_0\} \cup \{\emptyset\}) \\ &= E\text{-closure}(q_1 \cup q_0) \\ &= E\text{-closure}(q_0, q_1) \cup \{q_0, q_1, q_2\} \end{aligned}$$

$$S'(q_0, q) = \{q_0, q_1, q_2\}$$

$$\begin{aligned} S'(q_0, b) &= E\text{-closure}(\delta(E\text{-closure}(q_0), b)) \\ &= E\text{-closure}(\delta(q_0, q_1, q_2), b) \\ &= \text{" } (\delta(q_0, b) \cup \delta(q_1, b) \cup \delta(q_2, b)) \\ &= \text{" } (q_2 \cup q_1 \cup \{q_0, q_1\}) \\ &= \text{" } \{q_2 \cup q_1 \cup q_0\} \end{aligned}$$

$$S'(q_0, b) = \{q_0, q_1, q_2\}$$

$$\begin{aligned} S'(q_1, a) &= E\text{-closure}(\delta(E\text{-closure}(q_1), a)) \\ &= \text{" } (\delta(q_0, q_1), a)) \\ &= \text{" } (\delta(q_0, a) \cup \delta(q_1, a)) \\ &= \text{" } (\{q_0, q_1\} \cup \{q_0\}) \\ &= \text{" } (\{q_1 \cup q_0\}) \end{aligned}$$

$$S'(q_1, a) = \{q_0, q_1, q_2\}$$

$$\begin{aligned} S'(q_1, b) &= E\text{-closure}(\delta(E\text{-closure}(q_1), b)) \\ &= \text{" } (\delta(q_0, q_1), b)) \\ &= \text{" } (\delta(q_0, b) \cup \delta(q_1, b)) \\ &= \text{" } (q_2 \cup q_1) \end{aligned}$$

$$S'(q_1, b) = \{q_0, q_1, q_2\}$$

$$\begin{aligned} S'(q_2, a) &= E\text{-closure}(\delta(E\text{-closure}(q_2), a)) \\ &= \text{" } (\delta(q_0, q_1, q_2), a)) \\ &= \text{" } (\delta(q_0, a) \cup \delta(q_1, a) \cup \delta(q_2, a)) \\ &= \text{" } (\{q_0, q_1\} \cup \{q_0\} \cup \{\emptyset\}) \\ &= \text{" } \{q_0 \cup q_1\} \end{aligned}$$

$$S'(q_2, a) = \{q_0, q_1, q_2\}$$

$$\begin{aligned}
 S'(q_0, q_1, q_2, a) &= \epsilon\text{-closure}(\delta(\epsilon\text{-closure}(q_0, q_1, q_2), a)) \\
 &= \epsilon\text{-closure}(\delta(q_0, a) \cup \delta(q_1, a) \cup \delta(q_2, a)) \\
 &= \{q_0, q_1\} \cup \{q_0\} \cup \emptyset
 \end{aligned}$$

$$S'(q_0, q_1, q_2, a) = \{q_0, q_1, q_2\}$$

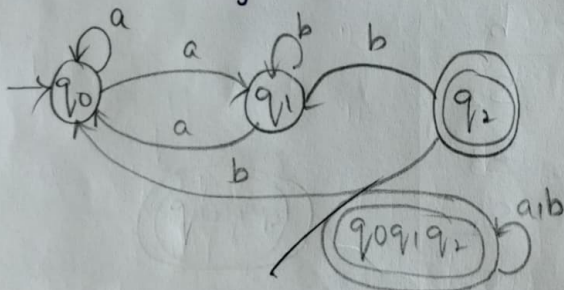
$$\begin{aligned}
 S'(q_0, q_1, q_2, b) &= \epsilon\text{-closure}(\delta(\epsilon\text{-closure}(q_0, q_1, q_2), b)) \\
 &= \epsilon\text{-closure}(\delta(q_0, b) \cup \delta(q_1, b) \cup \delta(q_2, b)) \\
 &= \{q_2\} \cup \{q_1\} \cup \{q_0, q_1\}
 \end{aligned}$$

$$S'(q_0, q_1, q_2, b) = \{q_0, q_1, q_2\}$$

Step 6: DFA Transition Table

State Input	a	b
→ q ₀	{q ₀ , q ₁ }	{q ₂ }
q ₁	{q ₀ }	{q ₁ }
* q ₂	∅	{q ₀ , q ₁ }
* q ₀ q ₁ q ₂	{q ₀ , q ₁ , q ₂ }	{q ₀ , q ₁ , q ₂ }

DFA state diagram:



(Q) Let L be the Language of words of length atleast 2 whose Last two letters are different from each other. For Example $abaab$ is in L , but not bb , construct an NFA for L , then draw the Equivalent DFA?

(A) Language: strings of length atleast 2 whose Last two symbols are different.

ex: Accepted

① ab

② ba

③ $aaab$

④ $abaab$.

Rejected

① aa

② bb

③ a

④ b .

NFA State

q_0 = start

q_1 = last symbol a

q_2 = last symbol b

q_3 = Final

Final state

* q_3

Equivalent DFA

S_0 = start

S_1 = Ends with a

S_2 = Ends with b

S_3 = Ends with ab (Accept)

S_4 = Ends with ba (Accept)

Transition table

State	a	b
→ S_0	S_1	S_2
S_1	S_1	S_3
S_2	S_4	S_2
* S_3	S_4	S_2
* S_4	S_1	S_3

Final state:

* S_3

* S_4

Transitions:

$q_0 \dots a \dots \rightarrow q_1$

$q_0 \dots b \dots \rightarrow q_2$

$q_1 \dots a \dots \rightarrow q_1$

$q_1 \dots b \dots \rightarrow q_3$

$q_2 \dots b \dots \rightarrow q_2$

$q_2 \dots a \dots \rightarrow q_3$

$q_3 \dots b \dots \rightarrow q_2$

$q_3 \dots a \dots \rightarrow q_1$

- (3) Design an NFA for binary strings whose last two bits are the same?

Language Ends with 00 or 11.

Formal definition:

$$M = (Q; E, q_0, S, F)$$

$$Q = \{q_0, q_1, q_2, q_3, q_4\}$$

$$\Sigma = \{0, 1\}$$

$$S = \{q_0\}$$

$$F = \{q_3, q_4\}$$

Transition Function

State	0	1
$\rightarrow q_0$	$\{q_0, q_1\}$	$\{q_0, q_2\}$
q_1	$\{q_3\}$	$\emptyset$
q_2	$\emptyset$	$\{q_4\}$
$* q_3$	$\emptyset$	$\emptyset$
$* q_4$	$\emptyset$	$\emptyset$

String:

$q_0 \rightarrow 0, q_0 q_1 \rightarrow 1, q_0 q_2 \rightarrow 0, q_0 q_1 \rightarrow 1, q_0 q_2 \rightarrow 1$

q_4 is final

Since q_4 is final

01011 is Accepted

4)

Construct DFA for strings containing 001 as a substring?

$$\Sigma = \{0, 1\}$$

Need substring : 001.

States: q_0 = No Match

q_1 = seen 0

q_2 = seen 00

q_3 = seen 001 (Final).

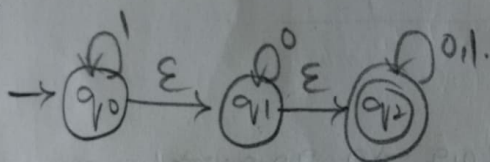
Transition Table:

state	0	1
$\rightarrow q_0$	q_1	q_0
q_1	q_2	q_0
q_2	q_2	q_3
$* q_3$	q_3	q_3

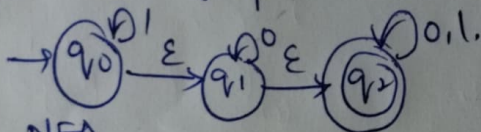
Final state:

q_3 is the DFA accepts ^{every} string Contains 001

5) A NFA with ϵ -transitions is given below. Find ϵ -closure for each state and design an Equivalent DFA with accepts the same language. Use its transition table mark initial and Final state?



Step 1: NFA ϵ -graph



Step 2: ^{NFA} Transition Table

State	ϵ	0	1
$\rightarrow q_0$	$\{q_1\}$	$\emptyset$	$\{q_0\}$
q_1	$\{q_2\}$	$\{q_1\}$	$\emptyset$
$* q_2$	$\emptyset$	$\{q_2\}$	$\{q_2\}$

Step 3:

$$\epsilon\text{-closure}(q_0) = \{q_0, q_1, q_2\}$$

$$\epsilon\text{-closure}(q_1) = \{q_1, q_2\}$$

$$\epsilon\text{-closure}(q_2) = \{q_2\}$$

step 4:

$$\begin{aligned}
 \delta'(q_0, 0) &= \epsilon\text{-closure}(\delta(\epsilon\text{-closure}(q_0, 0))) \\
 &= \epsilon\text{-closure}(\delta(q_0, q_1, q_2, 0)) \\
 &= \{ \delta(q_0, 0) \cup \delta(q_1, 0) \cup \delta(q_2, 0) \} \\
 &= \{ \emptyset \cup \{q_1\} \cup \{q_2\} \} \\
 &= \{q_1, q_2\} \\
 &= \{q_1, q_2\}
 \end{aligned}$$

$$\delta'(q_0, 1) = \{q_1, q_2\}$$

$$\begin{aligned}
 \delta'(q_0, 1) &= \epsilon\text{-closure}(\delta(\epsilon\text{-closure}(q_0, 1))) \\
 &= \{ \delta(q_0, 1, q_1, q_2) \} \\
 &= \{ \delta(q_0, 1) \cup \delta(q_1, 1) \cup \delta(q_2, 1) \} \\
 &= \{ q_0 \cup \emptyset \cup \{q_2\} \}
 \end{aligned}$$

$$\delta'(q_0, 1) = \{q_0, q_1, q_2\}$$

$$\begin{aligned}
 \delta'(q_1, 0) &= \epsilon\text{-closure}(\delta(\epsilon\text{-closure}(q_1, 0))) \\
 &= \{ \delta(q_1, 0, q_2, 1, 0) \}
 \end{aligned}$$

$$\begin{aligned}
 &= \{ \delta(q_1, 0) \cup \delta(q_2, 1, 0) \} \\
 &= \{ q_1 \cup q_2 \} = \{q_1, q_2\}
 \end{aligned}$$

$$\begin{aligned}
 \delta'(q_1, 1) &= \epsilon\text{-closure}(\delta(\epsilon\text{-closure}(q_1, 1))) \\
 &= \{ \delta(q_1, 1, q_2, 1, 1) \} \\
 &= \{ \delta(q_1, 1) \cup \delta(q_2, 1, 1) \} \\
 &= \{ q_2 \}
 \end{aligned}$$

$$\begin{aligned}
 \delta'(q_2, 0) &= \epsilon\text{-closure}(\delta(\epsilon\text{-closure}(q_2, 0))) \\
 &= \{ \delta(q_2, 0) \}
 \end{aligned}$$

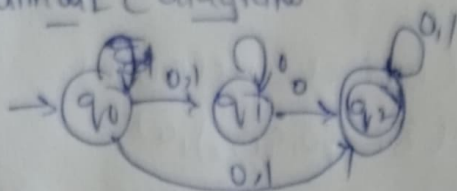
$$= \{q_2\}$$

$$\begin{aligned}
 \delta'(q_2, 1) &= \epsilon\text{-closure}(\delta(\epsilon\text{-closure}(q_2, 1))) \\
 &= \{q_2\}
 \end{aligned}$$

NFA with ϵ transition table

Slip	0	1
$\rightarrow q_0$	$\{q_1, q_2\}$	$\{q_0, q_1, q_2\}$
q_1	$\{q_1, q_2\}$	$\{q_2\}$
$* q_2$	$\{q_2\}$	$\{q_2\}$

NFA without ϵ diagram



Steps: $\delta(q_0, q_1, q_2, 0) = \delta(q_0, 0) \cup \delta(q_1, 0) \cup \delta(q_2, 0)$
 $= \{q_1, q_2\} \cup \{q_1, q_2\} \cup \{q_2\}$
 $= \{q_1, q_2\}$

$\delta(q_0, q_1, q_2, 1) = \delta(q_0, 1) \cup \delta(q_1, 1) \cup \delta(q_2, 1)$
 $= \{q_0, q_1, q_2\} \cup \{q_2\} \cup \{q_2\}$
 $= \{q_0, q_1, q_2\}$

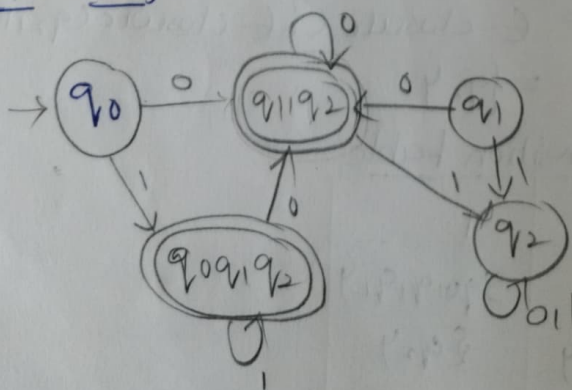
$\delta(q_1, q_2, 0) = \delta(q_1, 0) \cup \delta(q_2, 0)$
 $= \{q_1, q_2\} \cup \{q_2\}$
 $= \{q_1, q_2\}$

$\delta(q_1, q_2, 1) = \delta(q_1, 1) \cup \delta(q_2, 1)$
 $= \{q_2\} \cup \{q_2\}$
 $= \{q_2\}$

DFA T-table

Slip	0	1
$\rightarrow q_0$	$\{q_1, q_2\}$	$\{q_0, q_1, q_2\}$
q_1	$\{q_1, q_2\}$	$\{q_2\}$
$*q_2$	$\{q_2\}$	$\{q_2\}$
$*q_0, q_1, q_2$	$\{q_1, q_2\}$	$\{q_0, q_1, q_2\}$
$*q_1, q_2$	$\{q_1, q_2\}$	$\{q_2\}$

DFA - diagram



start state $\rightarrow q_0$

Final state:

$*q_1, q_2$

$*q_0, q_1, q_2$

$*q_2$

SIMATS ENGINEERING ASSESSMENT TOOL

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

91/100

31/07/2026